Xingtian (James) Si

Email: sixingtian@gmail.com | Mobile: +86 17757505168
Location: Zhejiang, China (open to relocate)

Research interests

Causal inference in economics; Causal ML (DML/TMLE/causal forests); Heterogeneous treatment effects; Policy evaluation & targeting (ads/coupons, education interventions).

EDUCATION

University of Nottingham Ningbo China Sep 2022 – Jul 2026
BSc Economics (Rank: Top 5%, expected first class) GPA: 3.88/4.0
Core courses: Microeconomics, Macroeconomics, Econometrics, Mathematical Economics, Python in Economics, Time Series, Real Analysis, Numerical Methods
Awards: Provost's Scholarship for Academic Excellence (Top 5%), UNNC (2023-2025), Zhejiang Provincial Scholarship (Oct 2025)

University of Nottingham Sep 2024 – Jan 2025
BSc Economics (Exchange Program) GPA: 4.0/4.0

RESEARCH EXPERIENCE

Dissertation: Causal ML and Counterfactual Inference in Consumer Behavior Sep 2025 – Present
Supervisor: **Zhenjiang Lin**, Associate Professor, University of Nottingham Ningbo China

- Estimate average and heterogeneous treatment effects of advertising/coupon interventions using **causal ML** (CRF/causal forests, **TMLE**, **BART**) in **R** (grf, tmle) and **Python** (scikit-learn).
- Build analysis-ready datasets from clickstream & purchase logs using **SQL/Python**; handle missingness via **MICE**; engineer covariates (demographics, purchase/browsing history).
- Develop a targeting / allocation workflow and communicate results with **R Shiny / Python Dash** dashboards (HTE profiles, counterfactual simulations).

Research Assistant: Early Childhood Intervention Effects (ECLS-K:2011) Mar 2025 – Sep 2025
Supervisor: **Jiaxi Yang**, PhD, Columbia University

- Clean and merge longitudinal **ECLS-K:2011** data (**n≈18,000**), apply sample weights, and impute missing values (MICE) in **Python/R**.
- Quantify baseline imbalance via standardized mean differences (Cohen's d) and variance ratios across multiple domains (demographic/academic/family, etc.).
- Train **CRF** to estimate **CATE/HTE** and build interpretable dashboards for subgroup results and policy simulations.

Empirical Project: Board Gender Diversity and Firm Performance Nov 2024 – Jun 2025
Supervisor: **Abigail Barr**, Professor, University of Nottingham

- Panel regressions on **840 firm-year** observations: OLS + year/industry controls; nonlinearity (quadratic terms) and interactions; heteroskedasticity checks (White test) in **Stata**.

PUBLICATIONS

Si, X. (2025). *Estimating Heterogeneous Treatment Effects of Early Childhood Interventions on Fifth-Grade Math Achievement: A Machine Learning-Augmented Causal Analysis of ECLS-K:2011.* **International Journal of Intelligence Science**, 15, 145–161.

Si, X., *Lin, K.*, Yang, X., *Wen, Z.* (2025). *Board Gender Diversity on Firm Performance in the UK: Evidence of Non-linear Returns and Firm Size Moderation (2008–2016).* **Entrepreneurship and Innovation**.

HONORS

Second Prize, "Huashu Cup" National Mathematical Modeling Competition (Aug 2025)

Excellent Award, National Tech Innovation & Entrepreneurship Competition (Sep 2025)
Second Prize, National College Student Mathematical Modeling Competition (Oct 2025)

SKILLS

Programming/Data: Python (pandas, numpy, scikit-learn), R (tidyverse, grf, tmle), Stata, SQL, MATLAB, LaTeX
Causal/metrics: HTE/CATE (causal forests/CRF), TMLE, BART; panel regressions; time series basics; dashboarding (R Shiny / Python Dash)